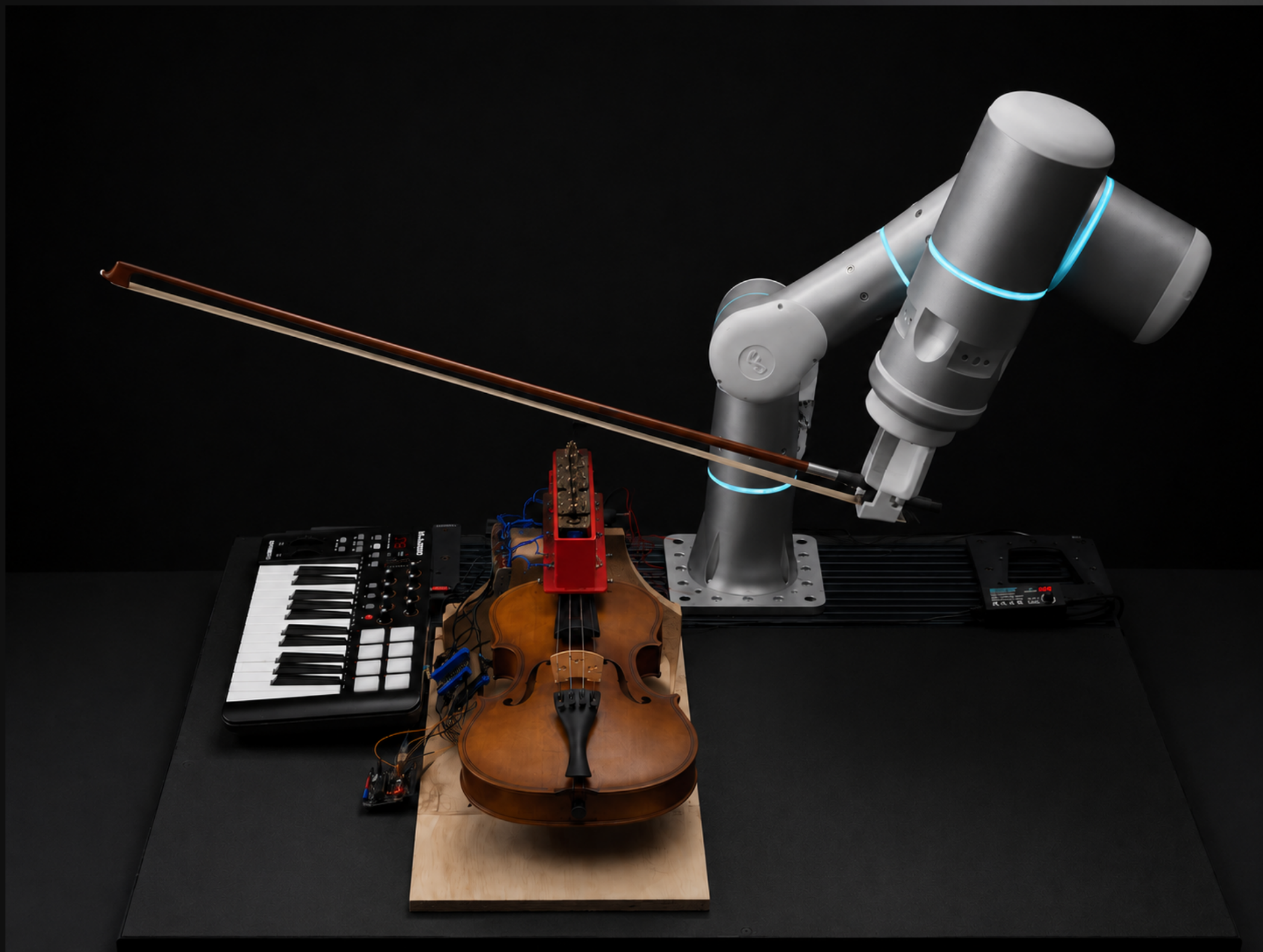


StradiBot

Team: Anna, Harold, Loïc, Joana

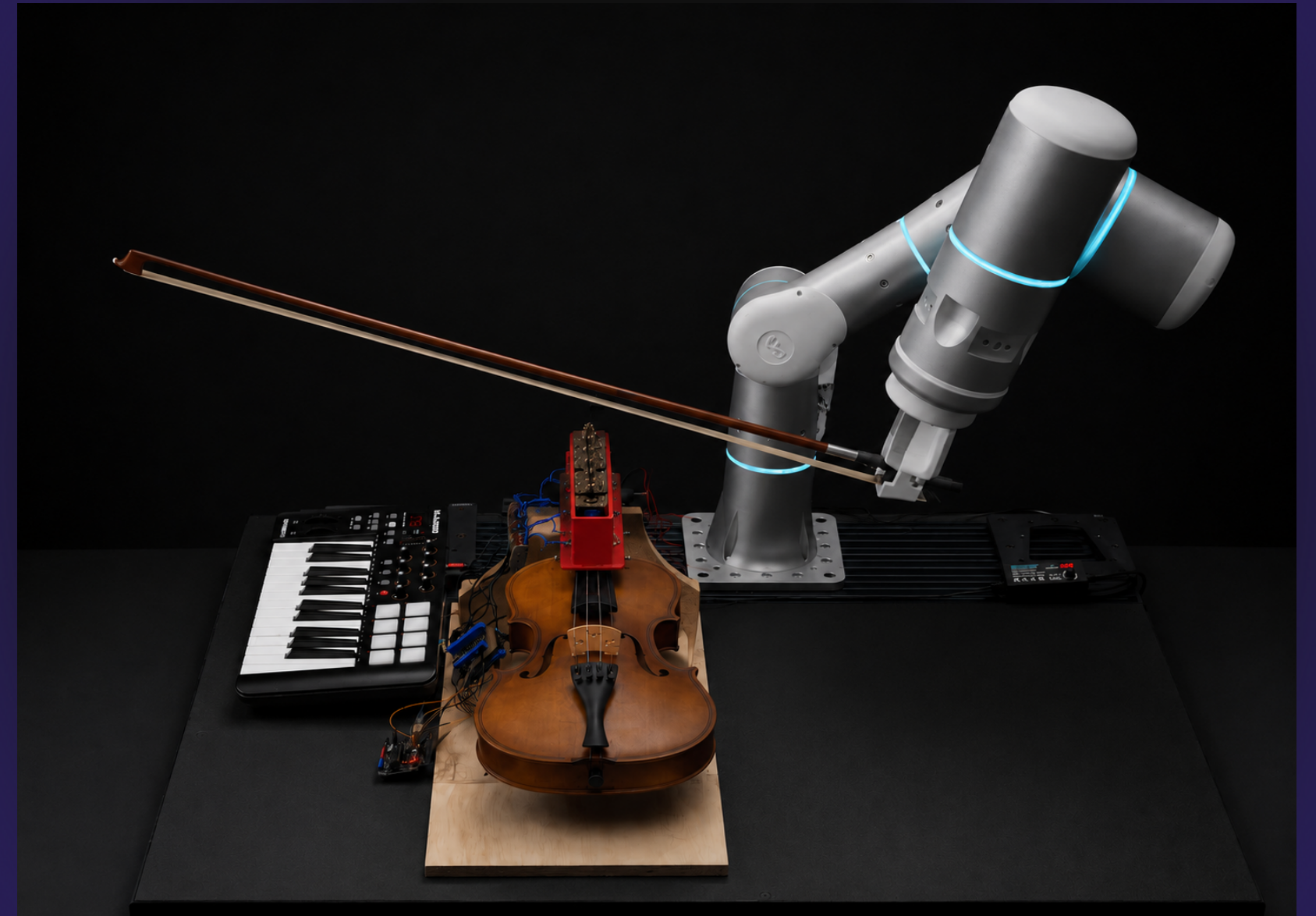
CS225A, Spring 2026



What is our project?

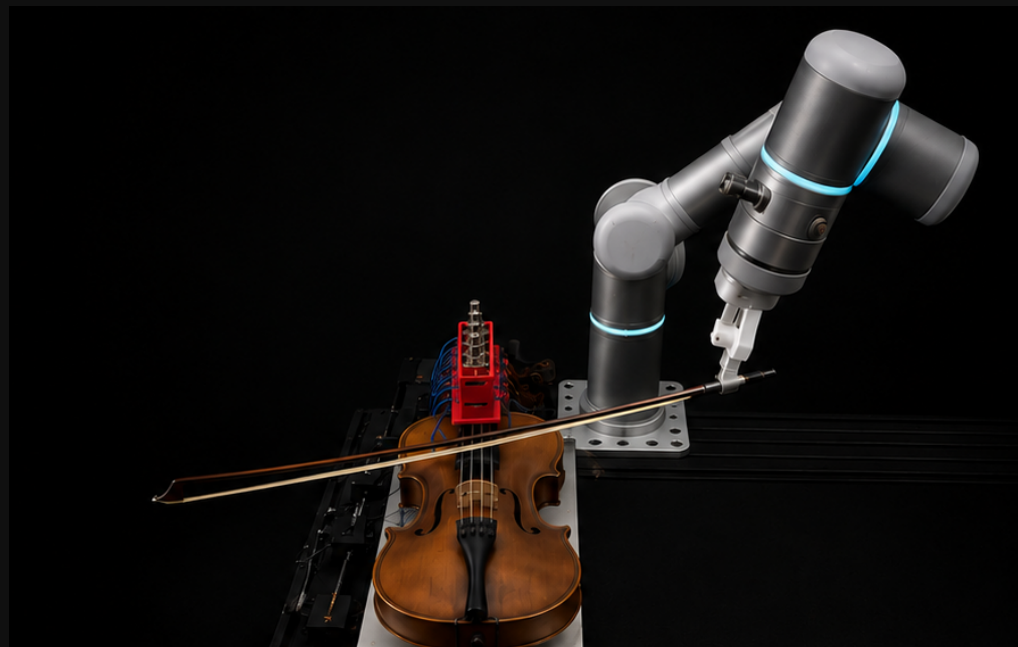
The goal

- A robotic arm performing violin bowing (Flexiv Rizon 4S, 7 DOF)
- An arduino mechanism for finger placement on the violin strings
- The output: recognizable musical notes from a MIDI file input or a MIDI keyboard



In more detail...

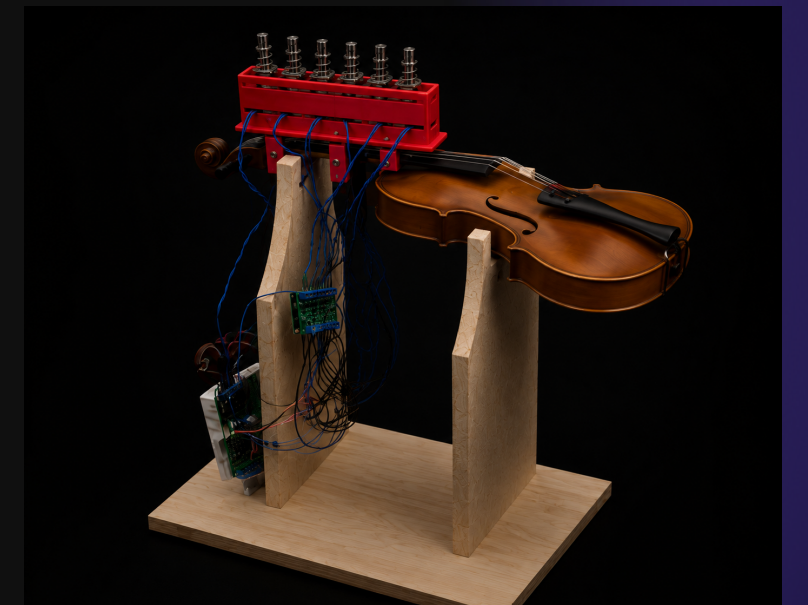
The bowing arm



The note pressing mechanism

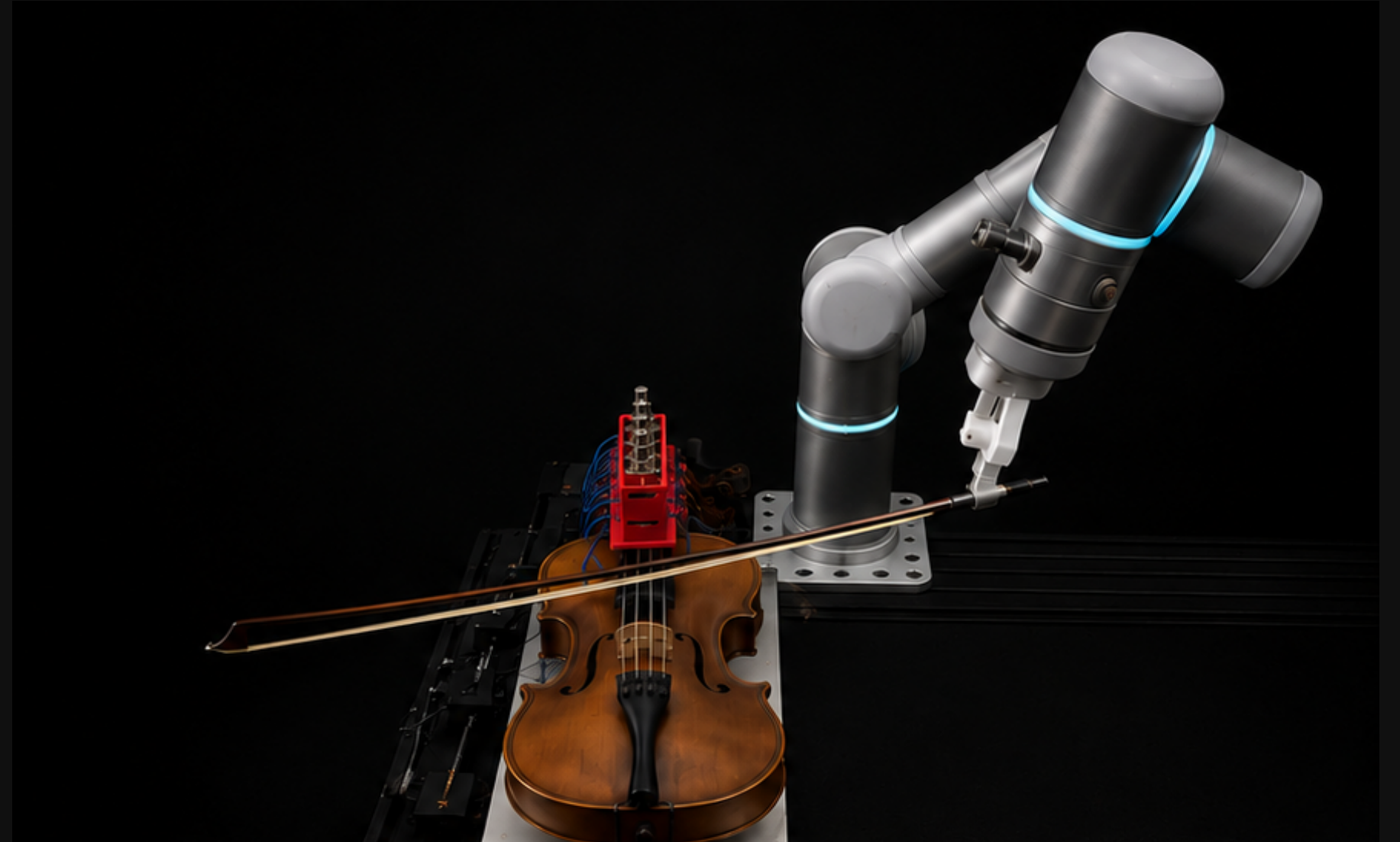


End-effector and stand design



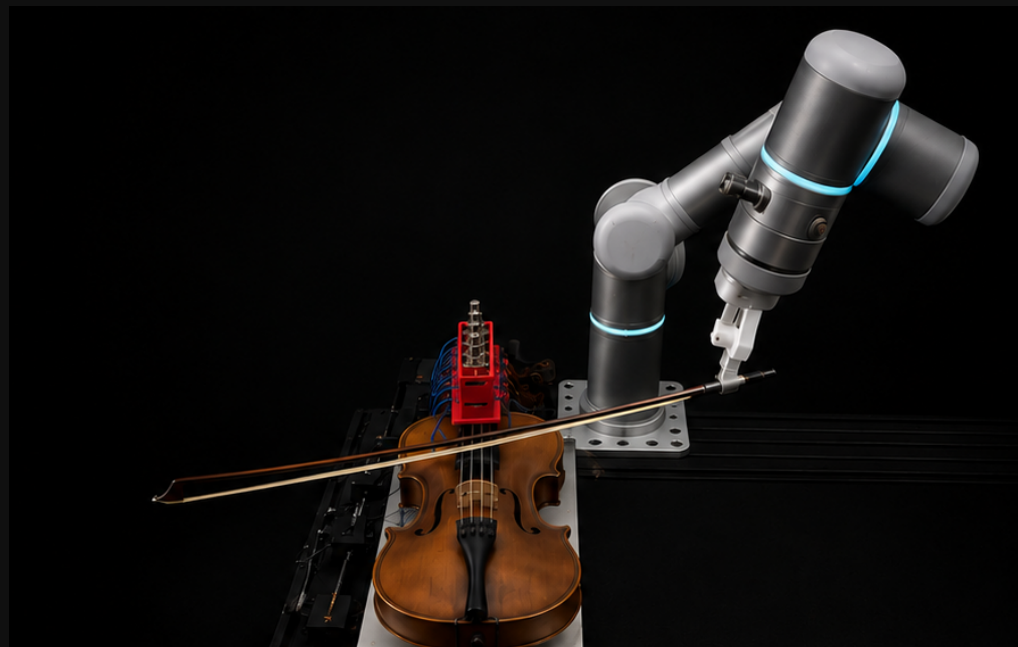
The bowing arm

- Flexiv 7-DOF: closed-loop force-controlled bow stroke
- Manual calibration for each string
- Automatic string changes (adjacent or not)



In more detail...

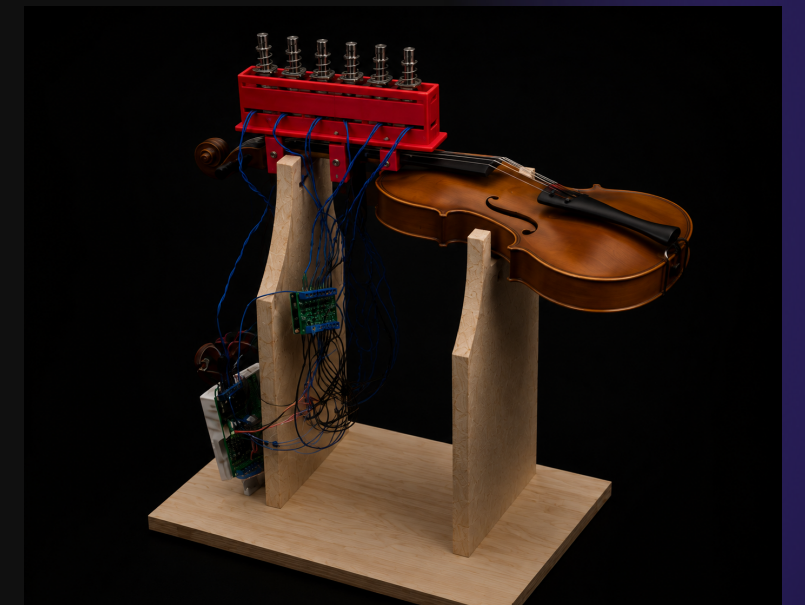
The bowing arm



The note pressing mechanism



End-effector and stand design



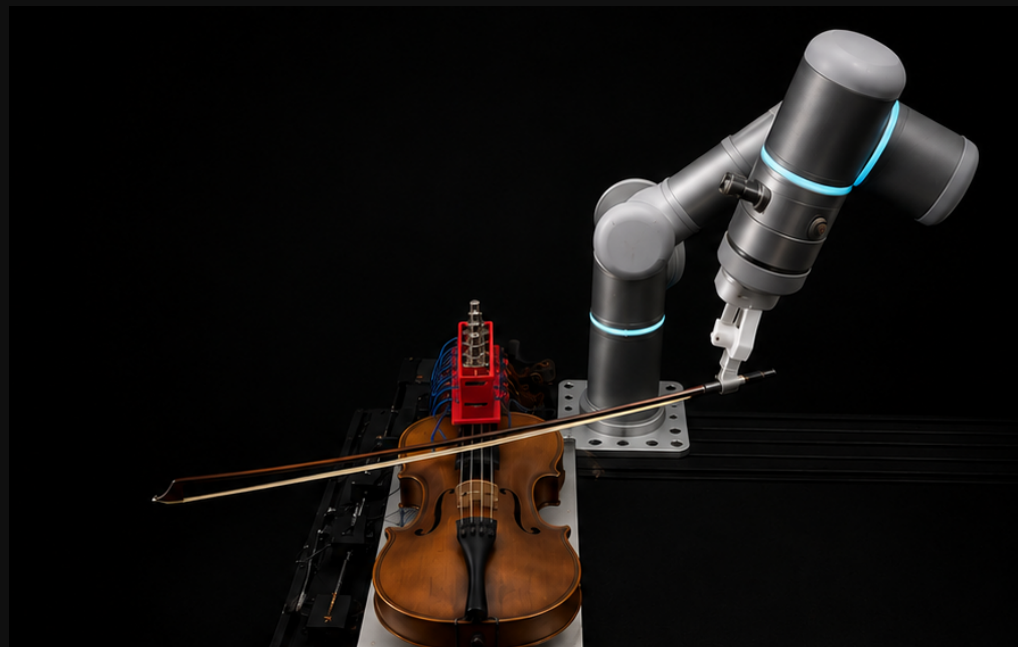
The note pressing mechanism

- Arduino Uno + MOFSET board + 6 solenoids along the fingerboard
- Each solenoid (24V, 42N) presses all 4 strings (only the bowed string produces sound)
- Design of mini EE for each solenoid to press down on the four strings



In more detail...

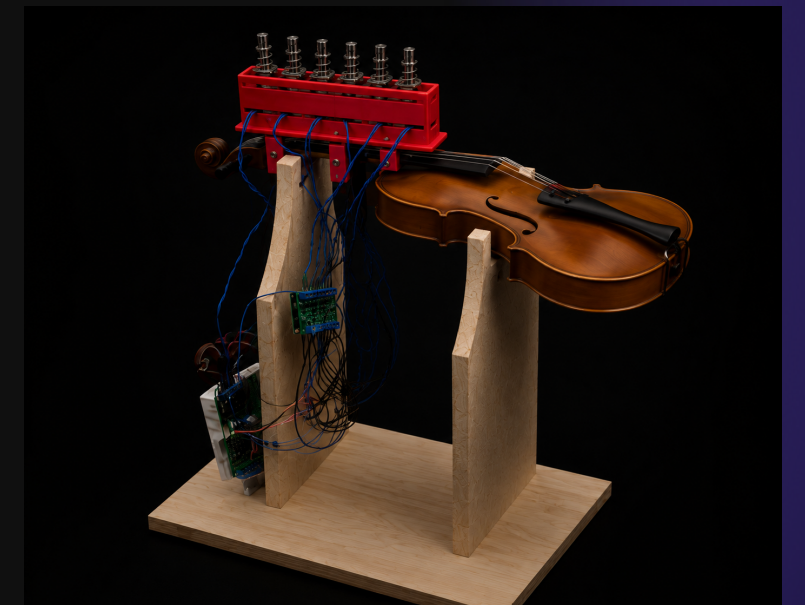
The bowing arm



The note pressing mechanism



End-effector and stand design

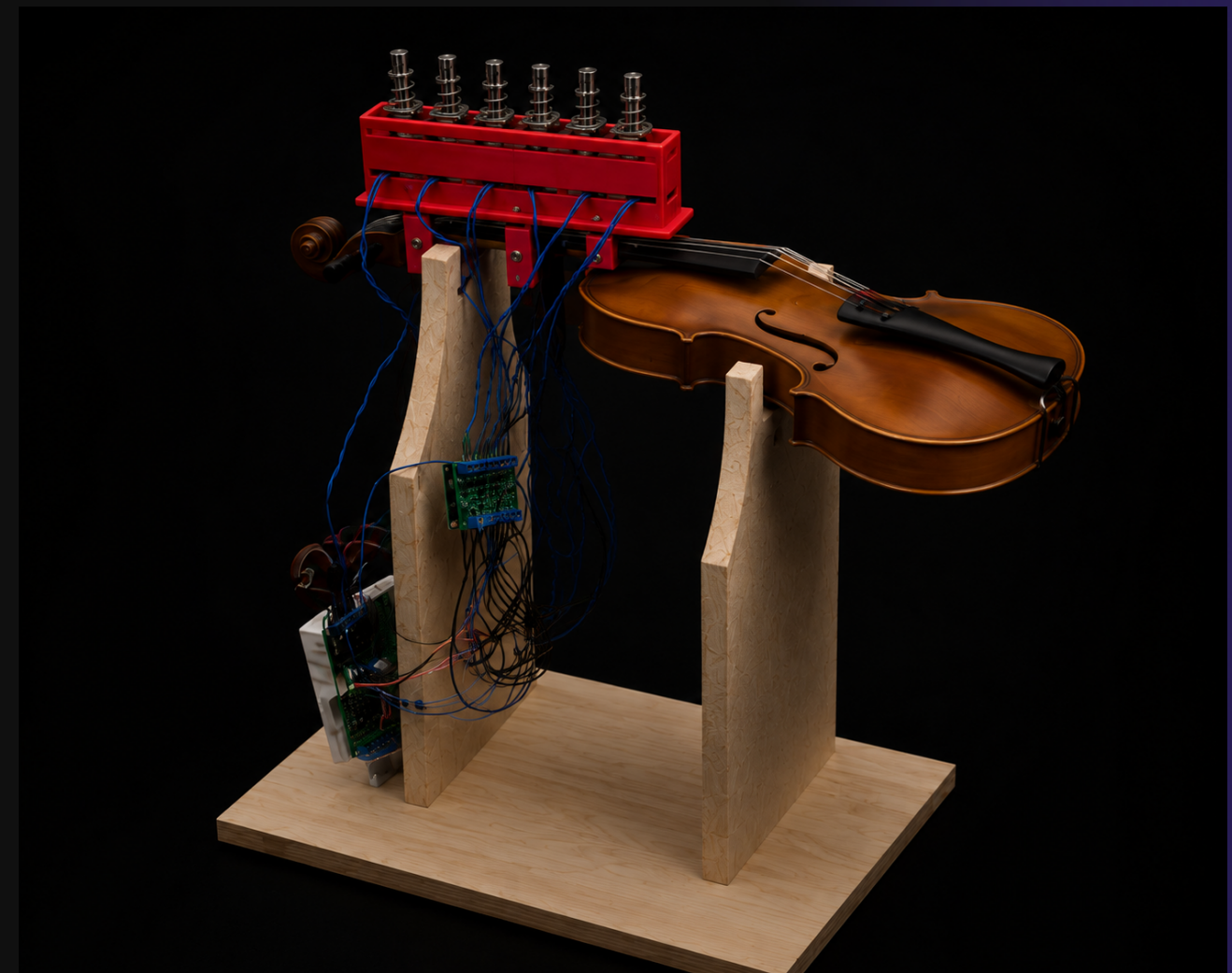


End-effector and stand design

- Custom 3D-printed bow holder mounted on the Flexiv



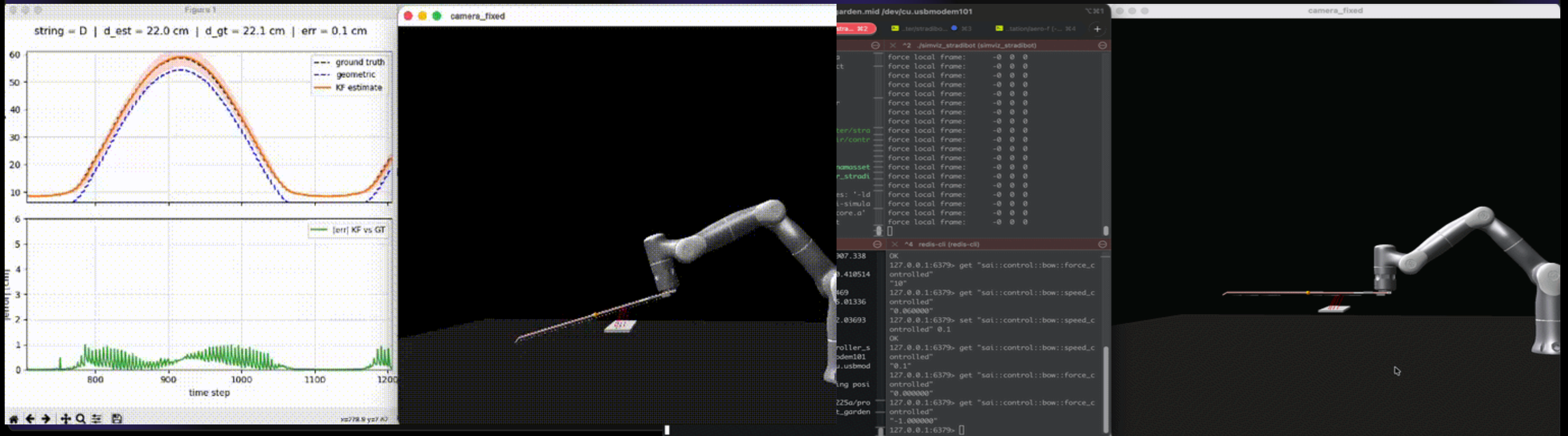
- Violin mounted on an elevated stand to provide clearance for the bow arm's range of motion during bowing strokes



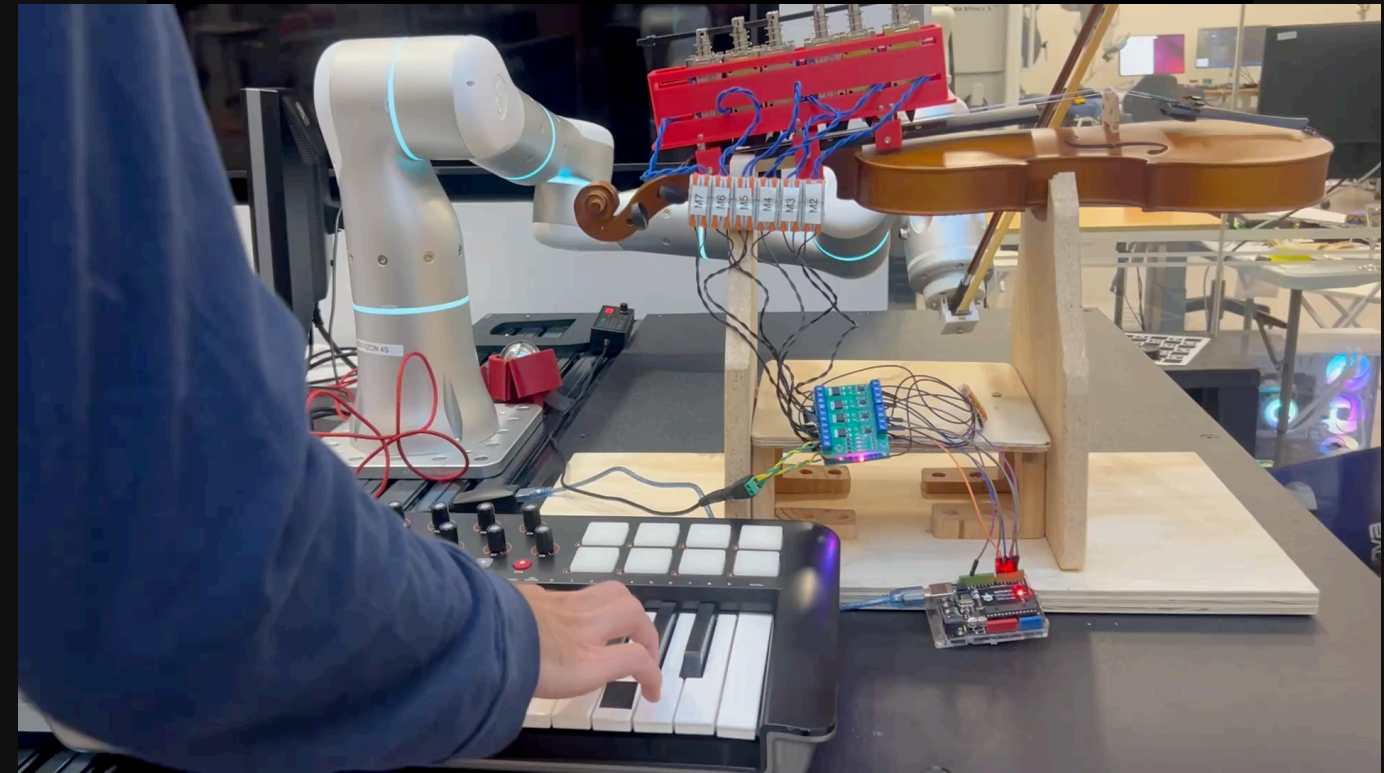
In simulation

In simulation

- OpenSai simulation with a Flexiv arm and a violin model (4 strings)
- Uses Redis for real-time communication between the FSM controller (Python) and the simulated robot
- Same FSM code runs on both simulated and real robot

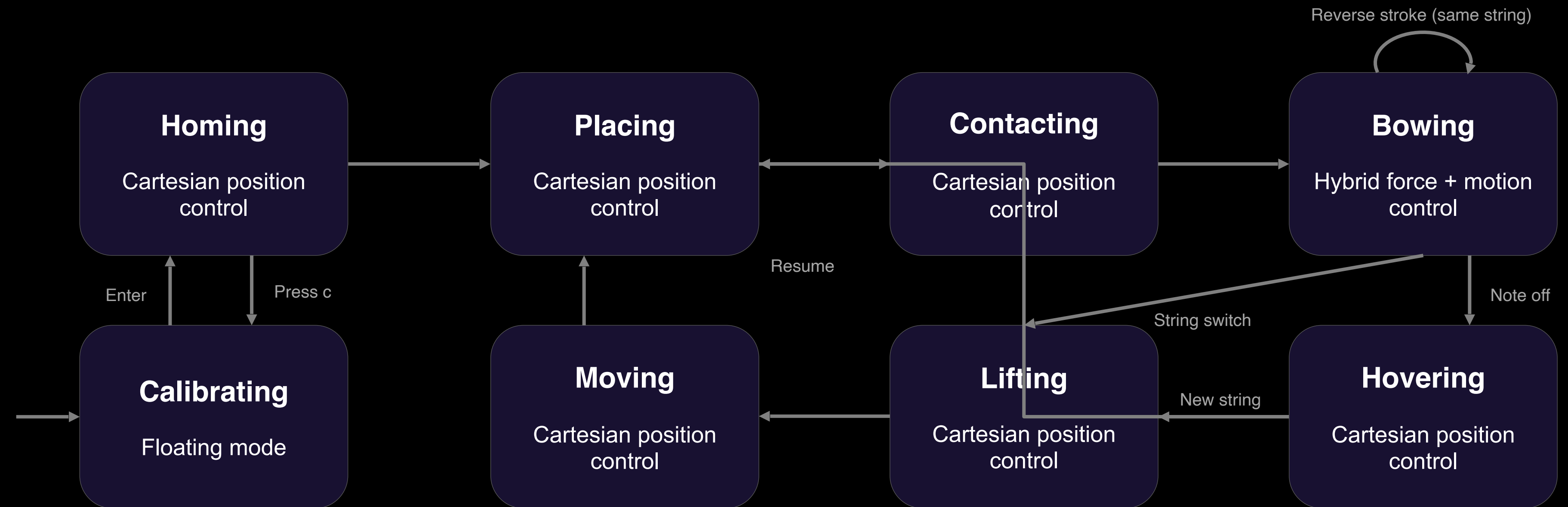


Human interface



1. **Keyboard controls:** Select strings (keys 5-8), start playback (p), recalibrate (c)
2. **MIDI file playback:** Loads a .mid file → robot parses note events and plays them in sequence
3. **Live MIDI keyboard:** Connect a USB MIDI keyboard (Oxygen 25) → plays notes in real time, robot maps each key to a string + fret combination and bows accordingly

Finite state machine



Press 0 from Bowing / Lifting / Hovering returns to Homing (safety reset). Calibrating can also load a saved calibration.

In the future?

Challenges

- Adjusting the force control gains
- Bow arm speed when switching between strings

Future work / improvement ideas

- String switching speed: faster trajectory planning for string transitions (optimize lift height and path to minimize time between notes)
- Integrate Kalman filter tracking the contact point into the FSM loop to make it more robust
- Automatic calibration using force sensing and vision instead of manual guide
- Dynamic bowing expression (varying speed and force)
- Vibrato using the solenoid or arm motion